



Multimodal Machine Learning

Lecture 4.1: Multimodal alignment

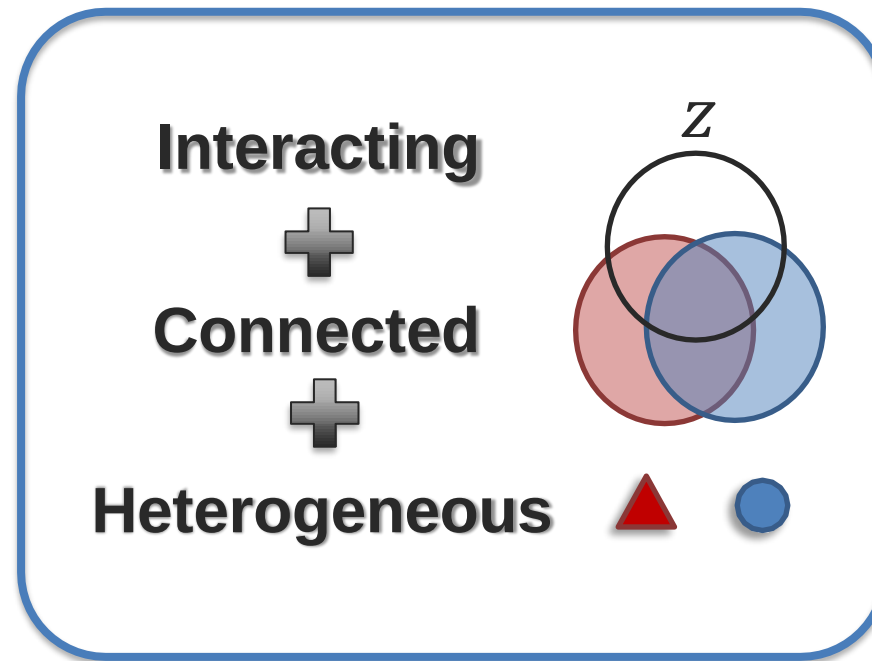
Dam Quang Tuan

Lecture objectives

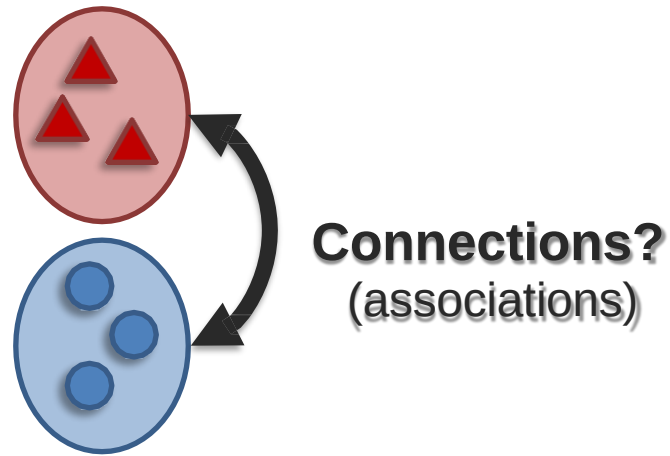
- A quick review
 - Connections, coordinated representations and mutual information
 - Modality interactions and factorized representations
- Discrete alignment
 - Local alignment
 - Coordinated representations; hard and soft attention
 - Global alignment
 - Assignment problem and optimal transport
- Continuous alignment
 - Continuous warping
 - Dynamic time warping
 - Discretization and segmentation

A Quick Review

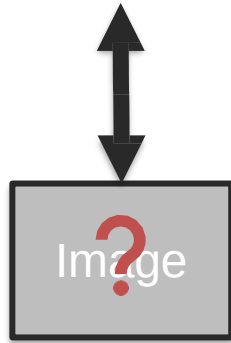
Multimodal is the scientific study of
heterogeneous, connected and interacting data



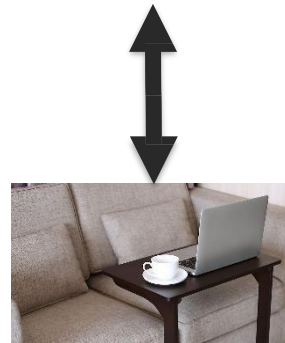
Connections between Modalities



"kitchen"



" ? "



Connection types:

●—● Co-occurrence

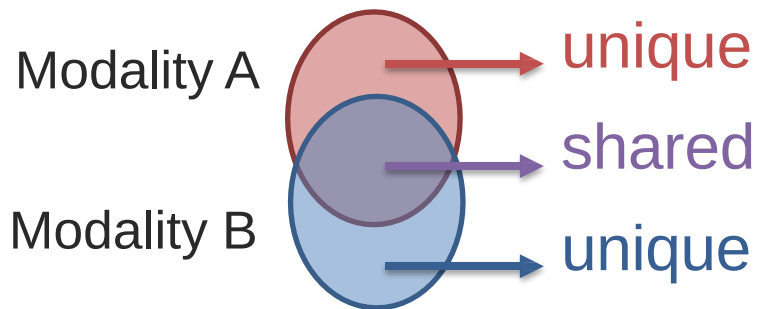
●—● Correlation

●—● Causality

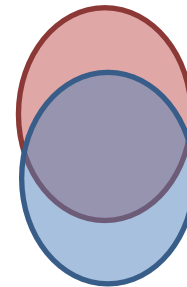
●—● Relationship

⋮

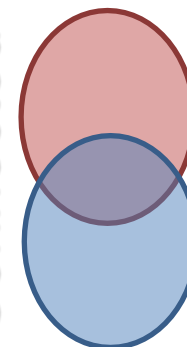
➔ knowledge of one modality provides information about the other modality



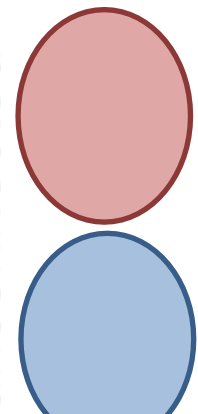
Strongly
Connected



Weakly
Connected

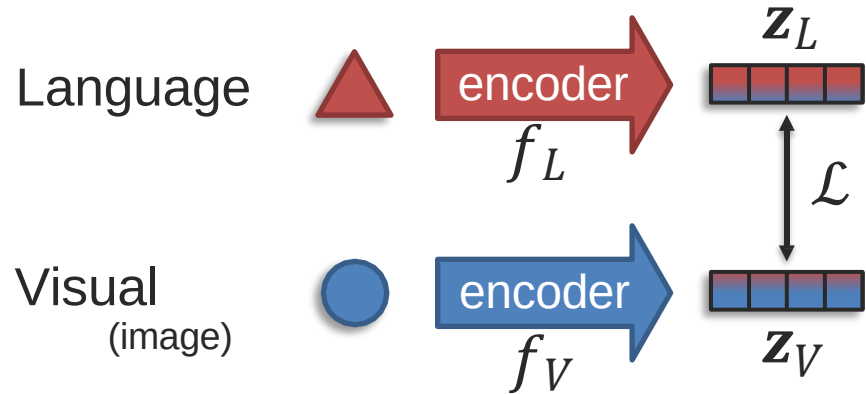


unconnected

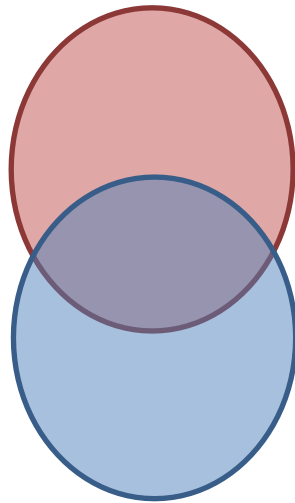
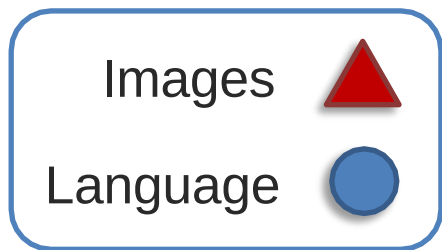


Connections are part of the data... and models will try to learn them.

Coordinated Representations – Example: CLIP



Connected modalities:



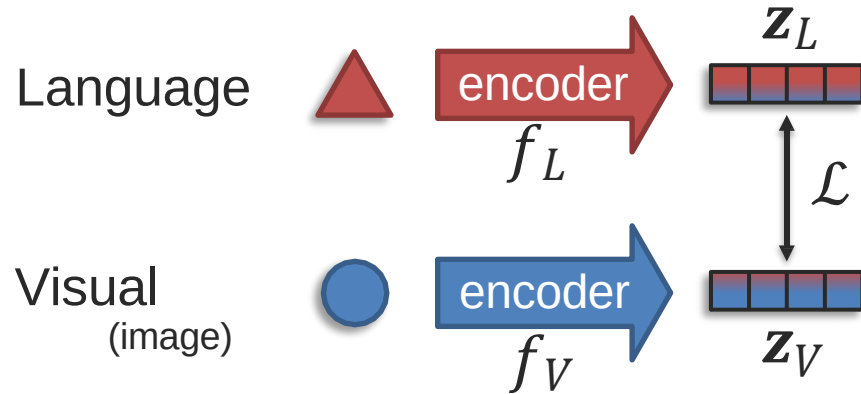
Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Annotations for the InfoNCE loss:

- $\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)$ (green box): positive pairs
- $\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)$ (red box): negative pairs and positive pairs
- Similarity function can be cosine similarity (orange text)

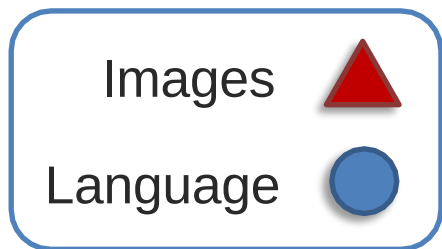
Coordinated Representations – Example: CLIP



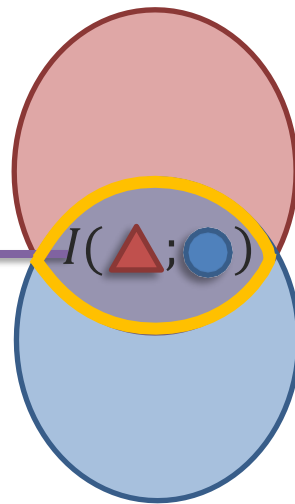
Popular contrastive loss: InfoNCE

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Connected modalities:



shared



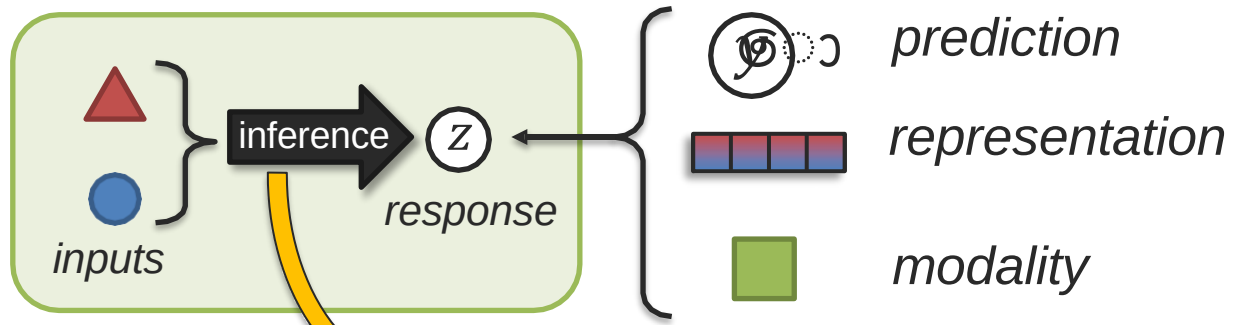
Mutual information $I(X; Y)$

$$\mathbb{E}_{X,Y} \left[\log \frac{P_{XY}(x, y)}{P_X(x)P_Y(y)} \right]$$



CLIP focuses on shared connections

Modality Interactions

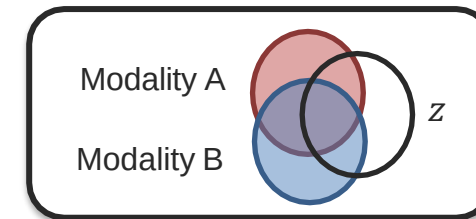


**Interactions happen
during inference!
(from human or model)**

Interactions require more
than the input modalities!

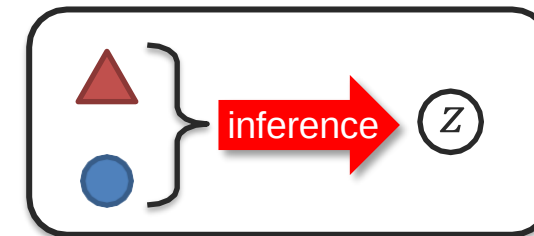
Interactions taxonomy:

Level 1: Response(s) and Input Modalities



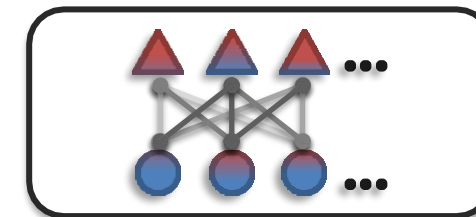
- ☐ Co-occurrence
- ☐ Redundancy
- ☐ Dominance
- ☐ Emergence
- ...

Level 2: Interactions – Internal Mechanics



- ☐ Additive
- ☐ Multiplicative
- ☐ Polynomial
- ☐ Nonlinear
- ...

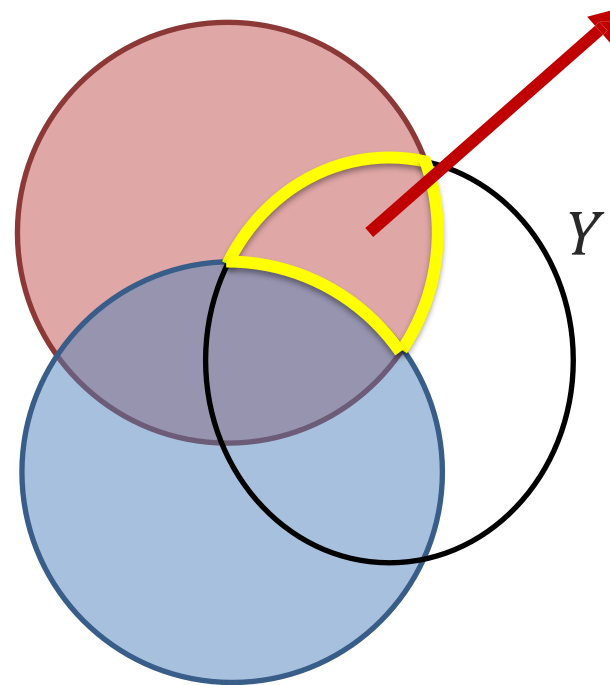
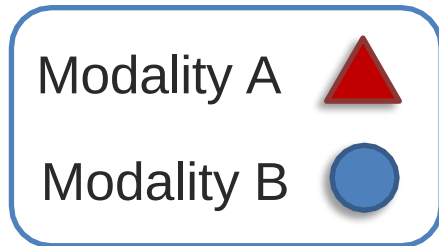
Level 3: Contextualized Interactions



- ☐ Temporal
- ☐ Hierarchy
- ☐ Multimodal
- ...

Modeling Interactions – Response and Input Modalities (Level 1)

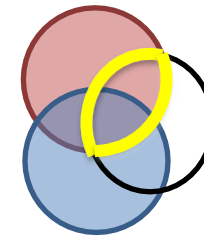
Information theory as a framework for modeling Level 1 interactions between input modalities and responses



Task-relevant **unique** information

$$I(\triangle; Y | \bullet)$$

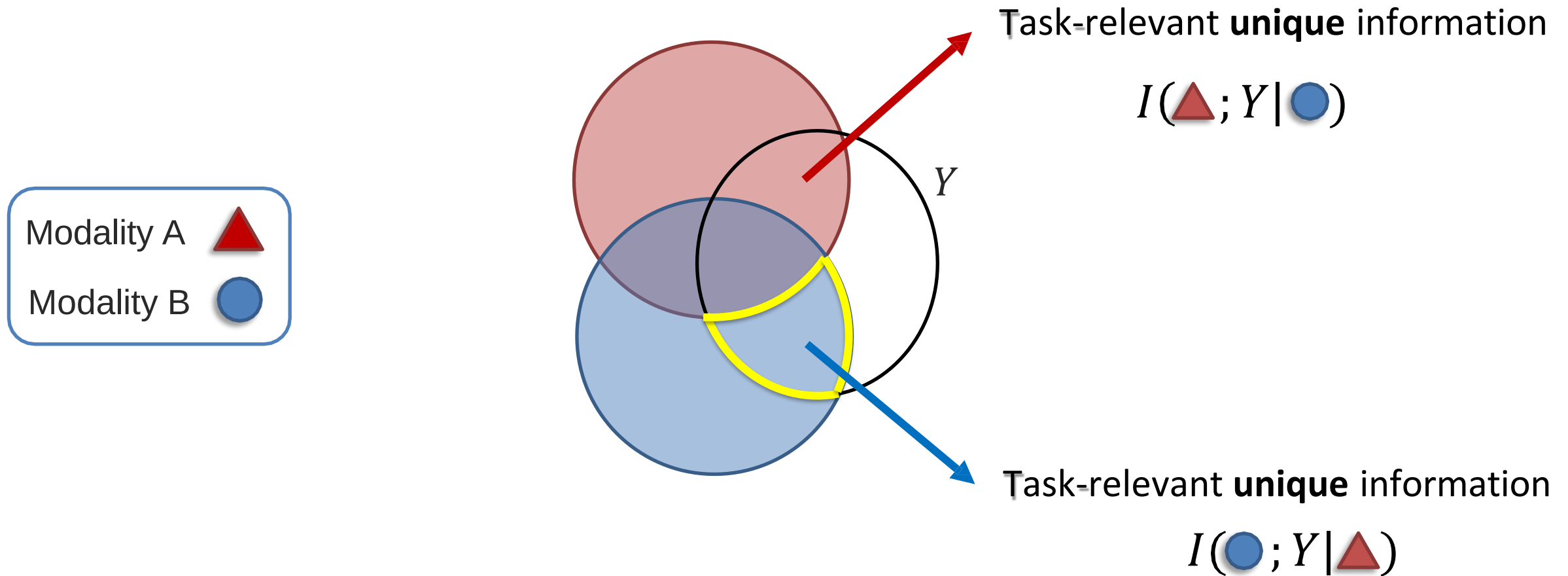
Mutual information...



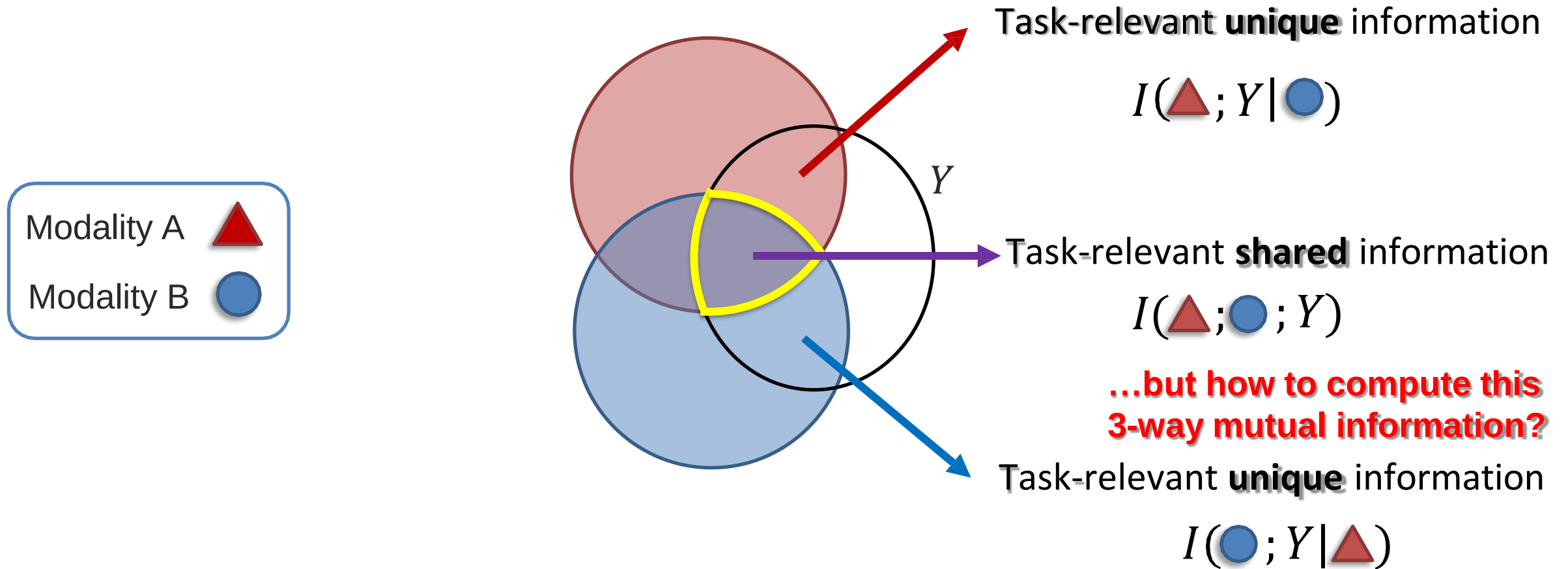
$$I(\triangle; Y)$$

...but without 

Modeling Interactions – Response and Input Modalities (Level 1)



Modeling Interactions – Response and Input Modalities (Level 1)



Modeling Interactions – Response and Input Modalities (Level 1)

Partial Information Decomposition (PID)

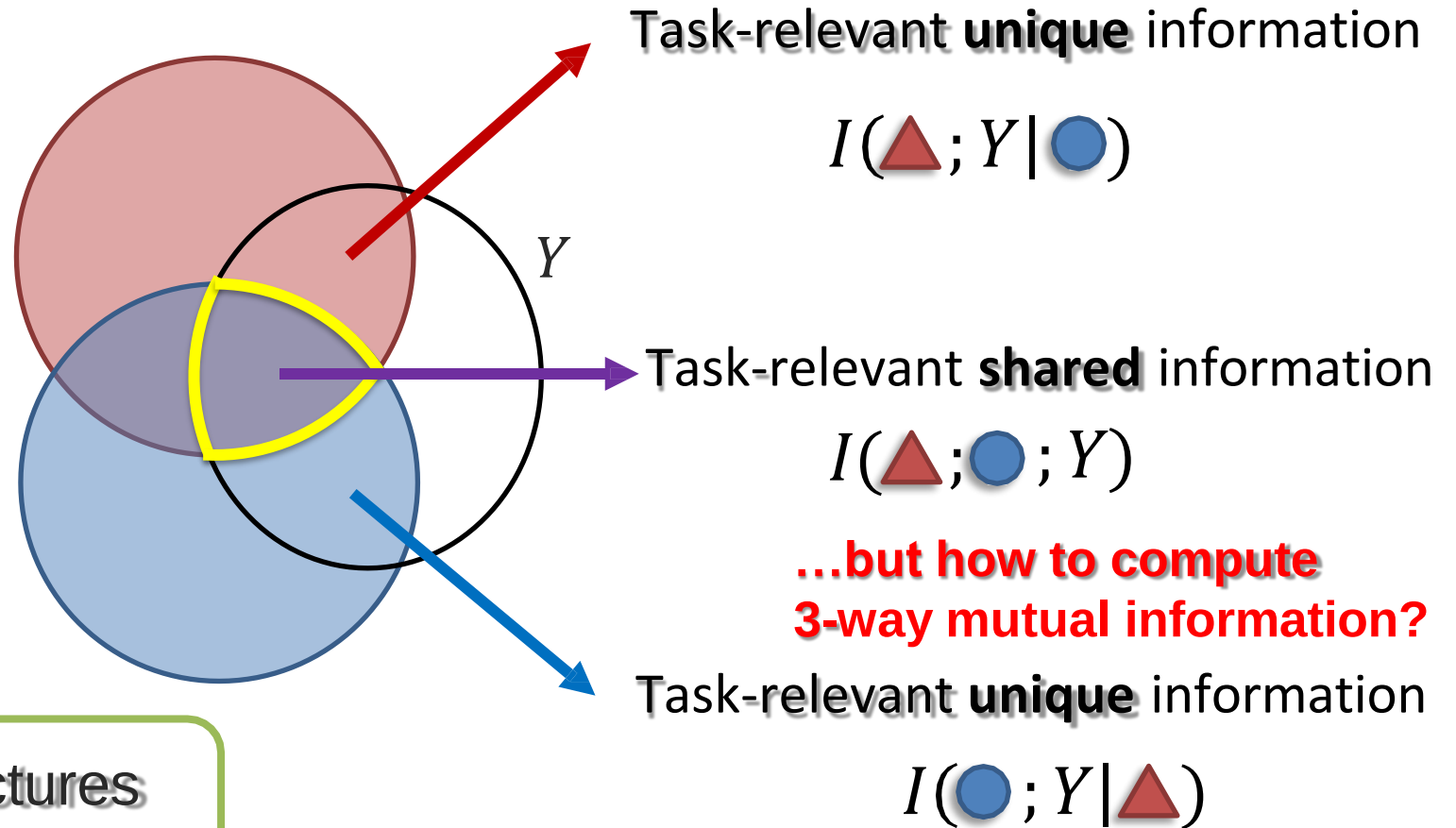
$$I(\triangle; \bullet; Y) = R - S$$

factorizes 3-way mutual information into:

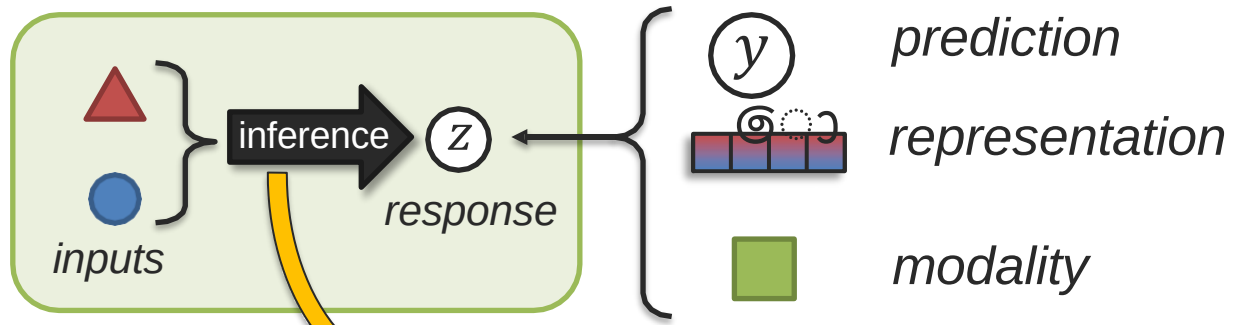
R: redundancy

S: Synergy

More about PID in future lectures
and reading assignments



Modality Interactions

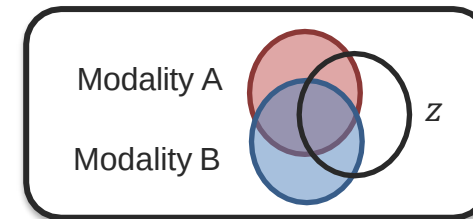


**Interactions happen
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(from human or model)**

Interactions require more
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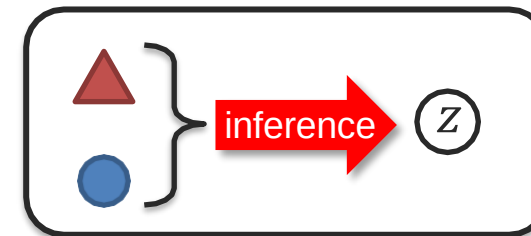
Interactions taxonomy:

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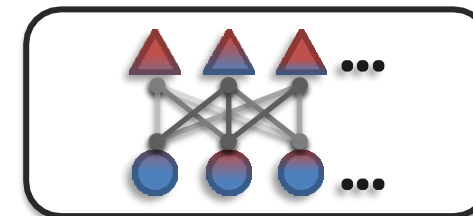
- ☐ Co-occurrence
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- ☐ Dominance
- ☐ Emergence
- ...

Level 2: Interactions – Internal Mechanics



- ☐ Additive
- ☐ Multiplicative
- ☐ Polynomial
- ☐ Nonlinear
- ...

Level 3: Contextualized Interactions



- ☐ Temporal
- ☐ Hierarchy
- ☐ Multimodal
- ...

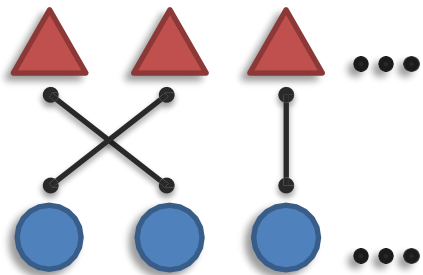
Challenge 2: Alignment

Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

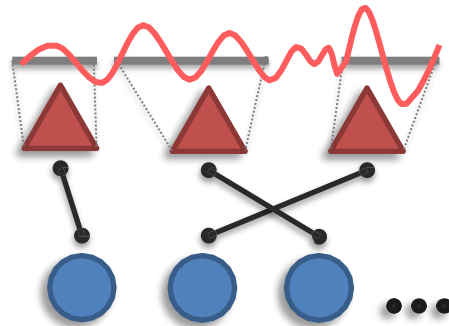
Sub-challenges:

Discrete Alignment



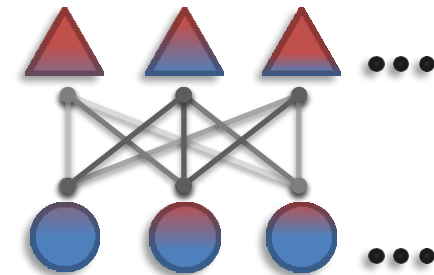
Discrete elements
and connections

Continuous Alignment



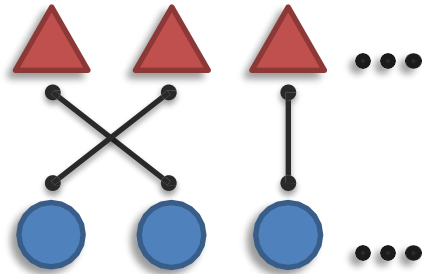
Segmentation and
continuous warping

Contextualized Representation

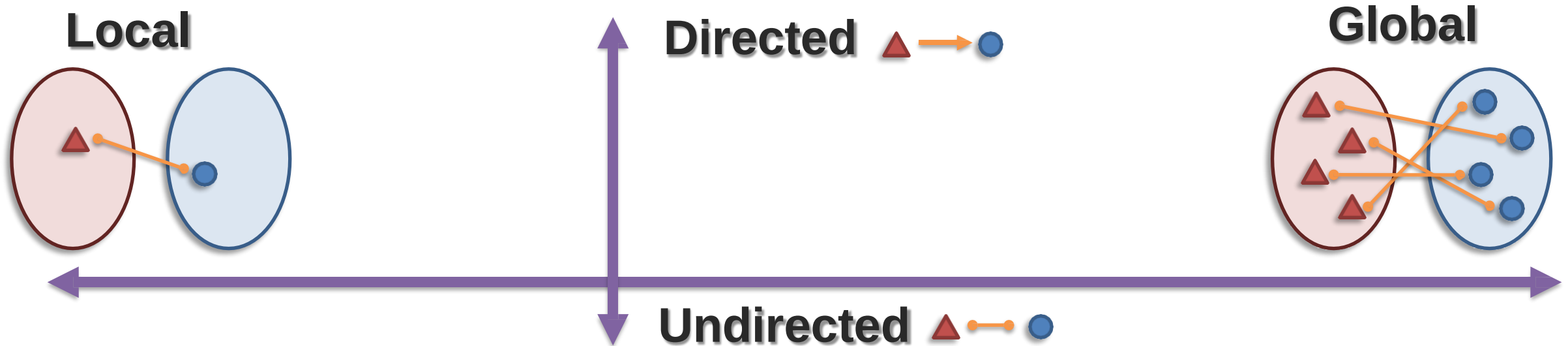


Alignment + representation

Sub-Challenge 2a: Discrete Alignment



Definition: Identify and model discrete connections between elements of multiple modalities



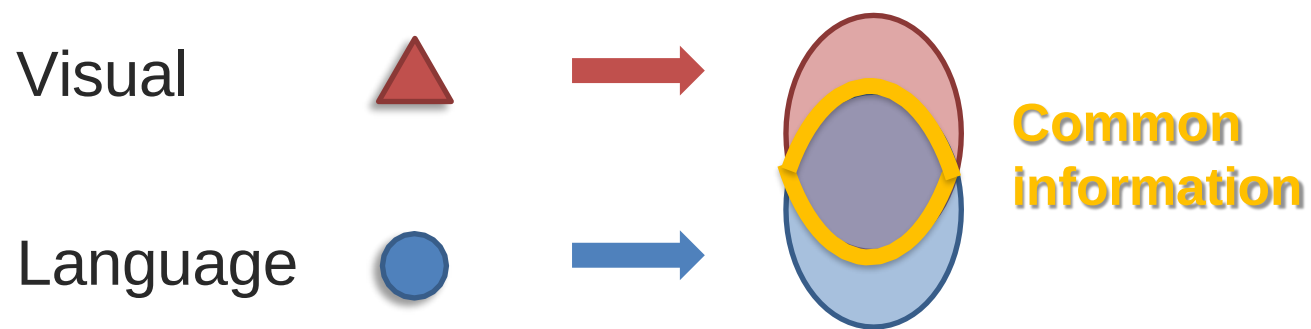
Language Grounding

Definition: Tying language (words, phrases,...) to non-linguistic elements, such as the visual world (objects, people, ...)



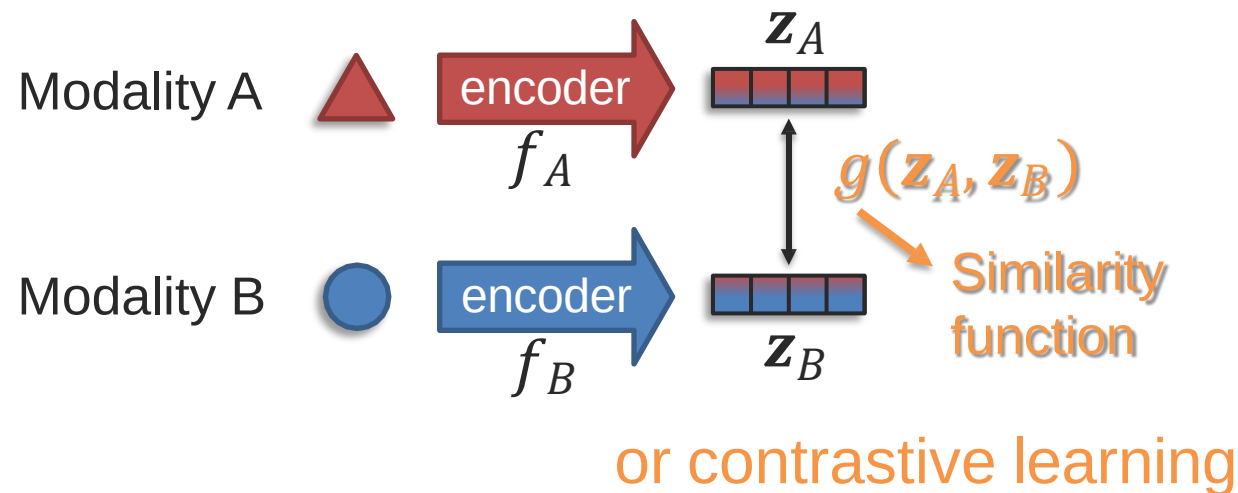
A **woman** reading **newspaper**

Local Alignment – Coordinated Representations

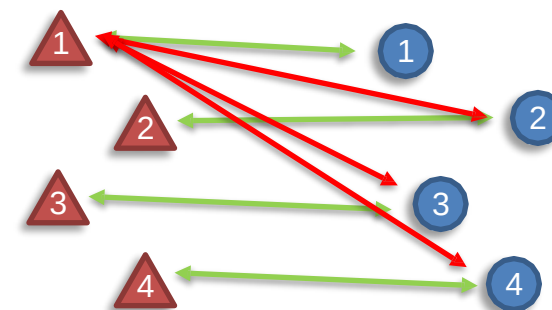


A **woman** reading **newspaper**

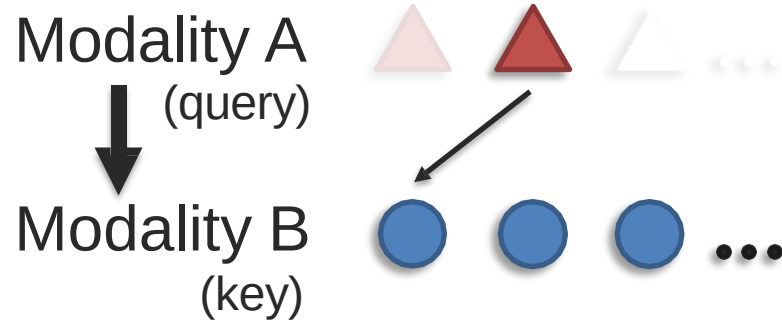
Learning coordinated representations:



Supervision: Paired data



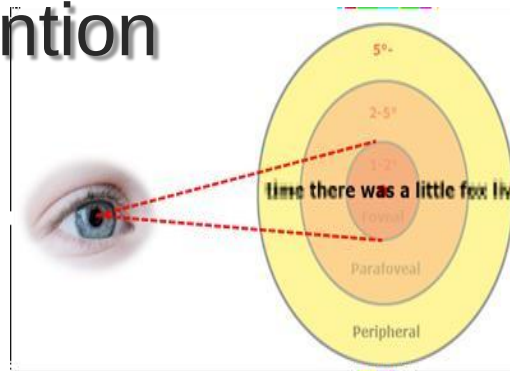
Directed Alignment



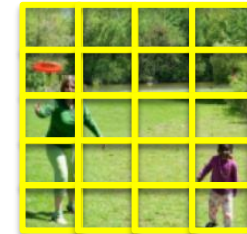
A woman is throwing a frisbee

Which
object?

Attention



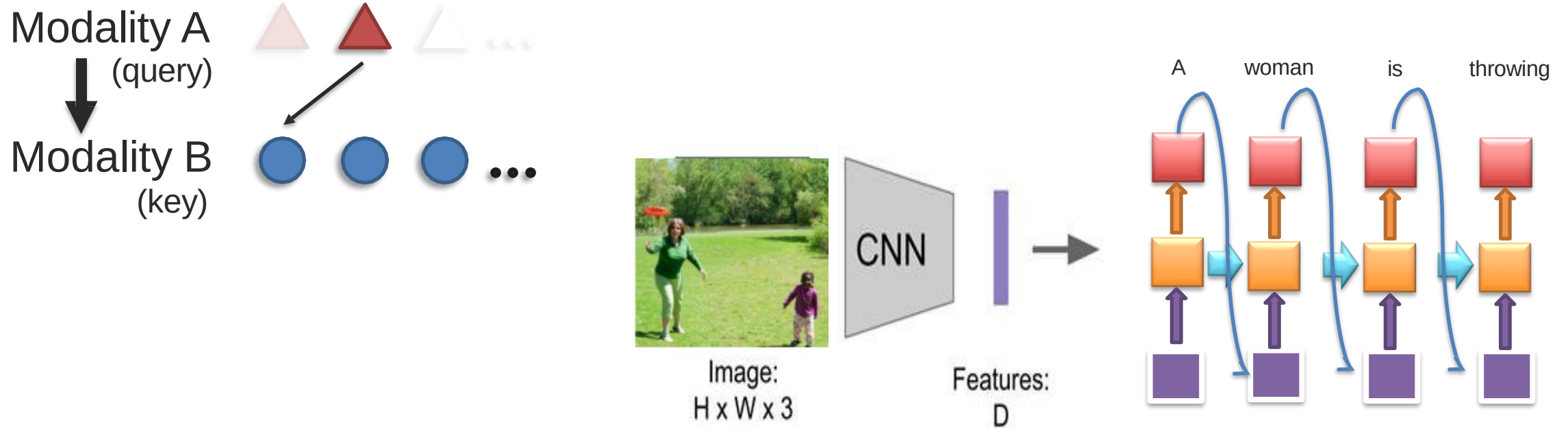
1 Soft attention



2 Hard attention

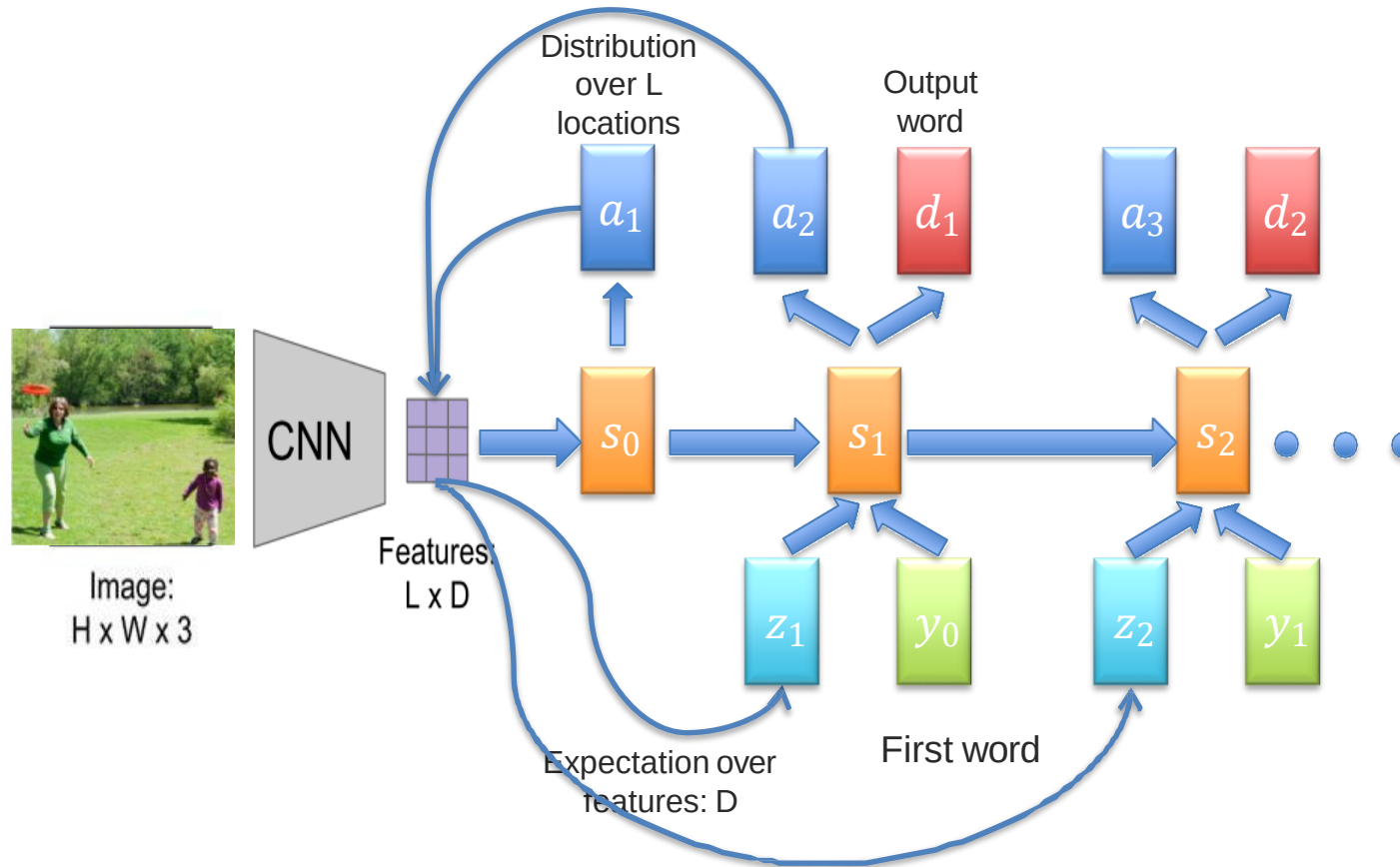


Directed Alignment – Image Captioning



Should we always use the final layer of the CNN for all generated words?

Directed Alignment – Image Captioning



Attention Gates

Before:

$$p(y_i|y_1, \dots, y_{i-1}, \mathbf{x}) = g(y_{i-1}, \mathbf{s}_i, \mathbf{z}),$$

where $\mathbf{z} = \mathbf{h}_T$, last encoder state and $\mathbf{s}_i$ is the current state of the decoder

Now:

$$p(y_i|y_1, \dots, y_{i-1}, \mathbf{x}) = g(y_{i-1}, \mathbf{s}_i, \mathbf{z}_i)$$

Have an attention “gate”

- A different context $\mathbf{z}_i$ used at each time step!
- $\mathbf{z}_i = \sum_{j=1}^{T_x} \alpha_{ij} \mathbf{h}_j$

α_{ij} is the (scalar) attention for word j at generation step i

Attention Gates

So how do we determine α_{ij} ?

$$\alpha_{i,j} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})} \quad \Rightarrow \text{softmax, making sure they sum to 1}$$

where:

$$e_{ij} = \mathbf{v}^T \sigma(W \mathbf{s}_{i-1} + U \mathbf{h}_j)$$

a feedforward network that can tell us how important the current encoding is

$\mathbf{v}$, W , U – learnable weights

$$\mathbf{z}_i = \sum_{j=1}^{T_x} \alpha_{ij} \mathbf{h}_j$$

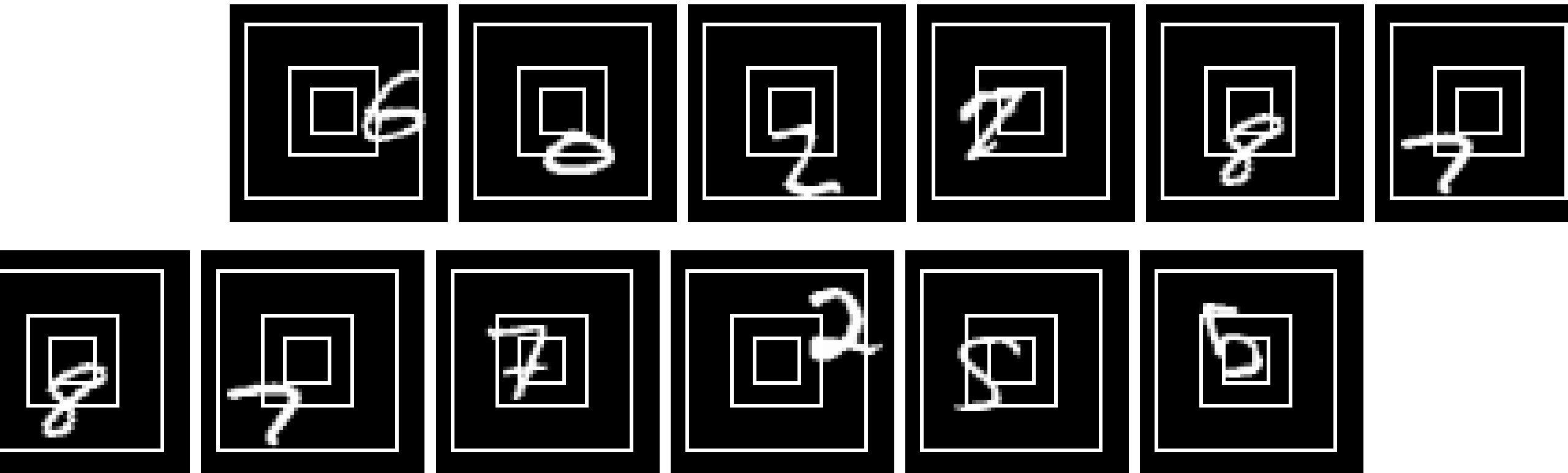
← expectation of the context (a fancy way to say it's a weighted average)

Example – Image Captioning

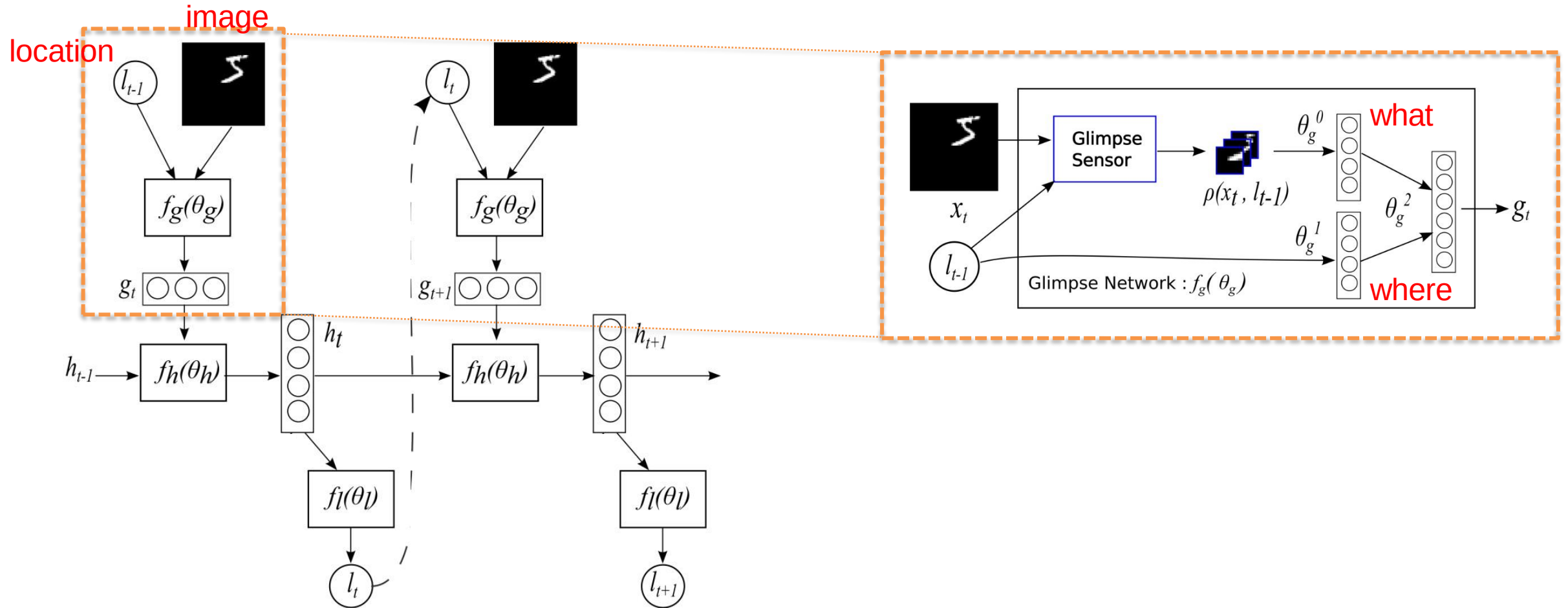


[Show, Attend and Tell: Neural Image Caption Generation with Visual Attention, Xu et al., 2015]

Hard attention - Example



Hard Attention – Recurrent Model of Visual Attention



Global Alignment

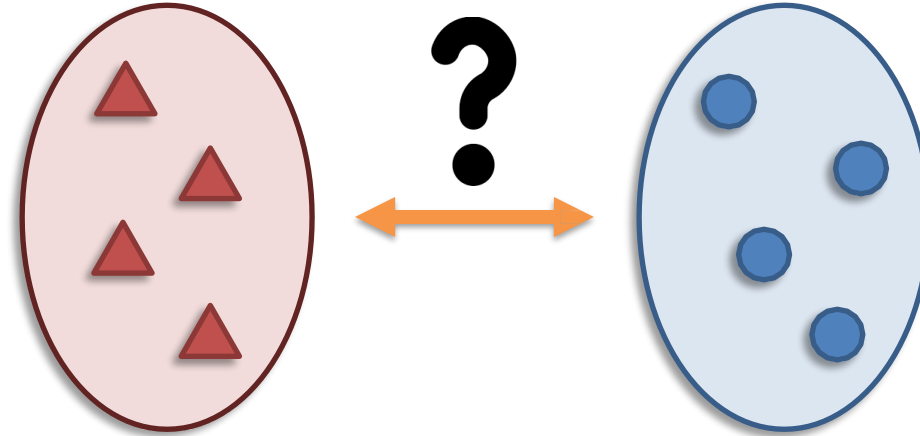
Visual



Language



Latent pairing information



Jointly optimize representation + global alignment:

Modality A



encoder

f_A



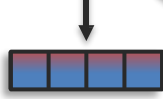
z_A

Modality B



encoder

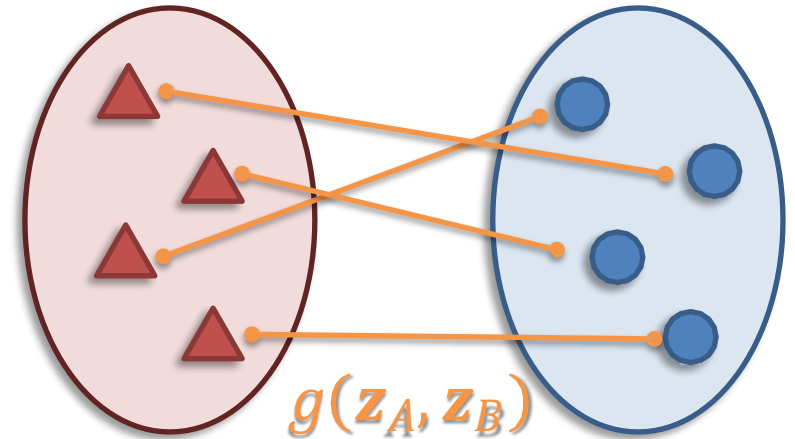
f_B



z_B

$g(z_A, z_B)$

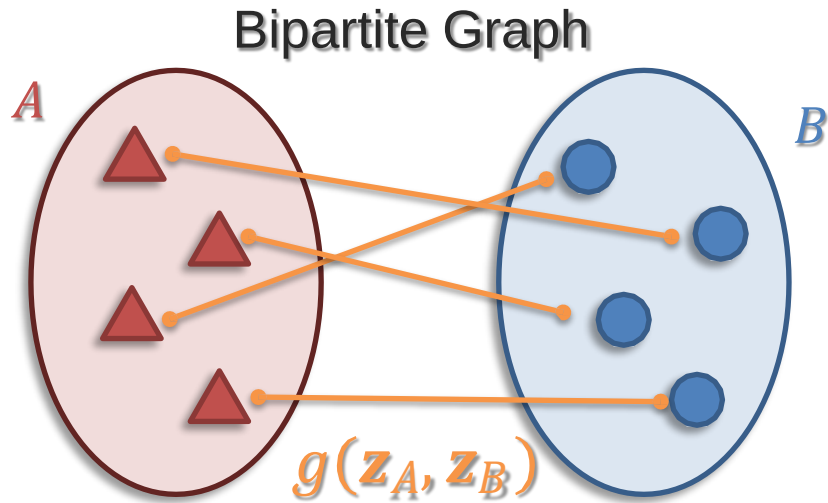
Coordination
function



(representation)

(global alignment)

Assignment Problem



Initial assumptions:

- Same number of elements in A and B modalities
- 1-to-1 “hard” alignment between elements
- All elements assigned (aka “perfect matching”)

➡ How to solve?

Naive solution: check all assignments

Better solution: Linear Programming

Assignment: ~~$f: A \rightarrow B$~~
(vector of indices)

Similarity weights: ~~$w_{(i, f(i))} = g(\mathbf{z}_A^i, \mathbf{z}_B^{f(i)})$~~

Maximize: ~~$\max_{f \in \text{Perm}(N)} \sum_{i=1}^N w_{i, f(i)}$~~

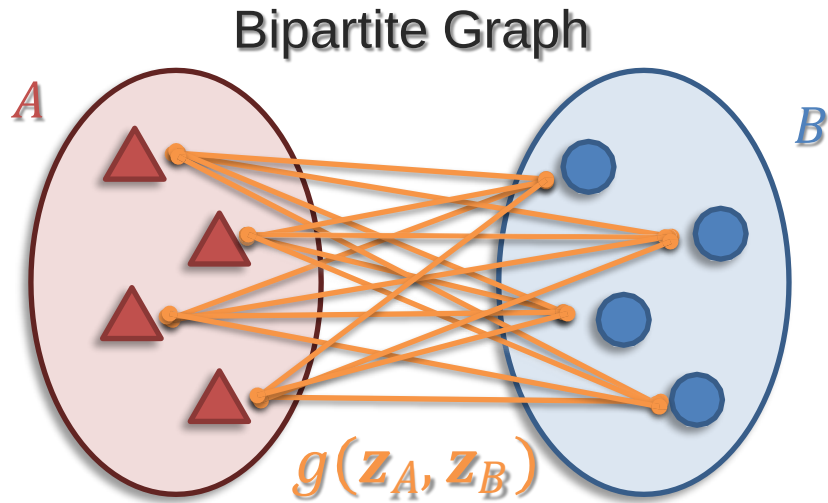
$x_{ij} = 1$ when matching connection, otherwise 0

$w_{(i,j)} = g(\mathbf{z}_A^i, \mathbf{z}_B^j)$

$\max_{\{x_{ij}\}} \sum_{(i,j) \in A \times B} w_{i,j} x_{ij}$

➡ Can be solved with
simplex algorithm

Optimal transport



New assumptions:

- Different number of elements in A and B modalities
- Many-to-many “soft” alignment between elements

It can be seen as “transporting” elements from modality A to modality B (and vice-versa)

Assignments: $x_{(i,j)}$: soft alignment between $\mathbf{z}_A^i$ and $\mathbf{z}_B^j$

Similarity weights: $w_{(i,j)} = g(\mathbf{z}_A^i, \mathbf{z}_B^j)$

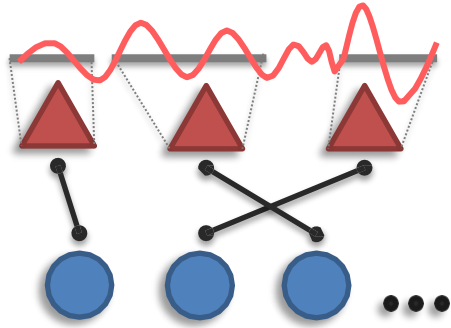
Maximize: $\max_{\{x_{ij}\}} \sum_{(i,j) \in A \times B} w_{i,j} x_{ij}$



Wassertein distance
gives optimal transport

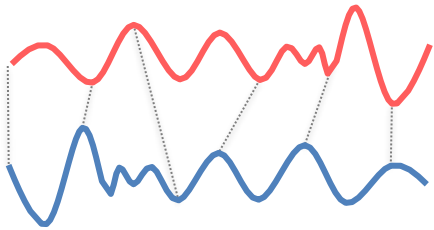
Continuous Alignment

Challenge 2b: Continuous Alignment

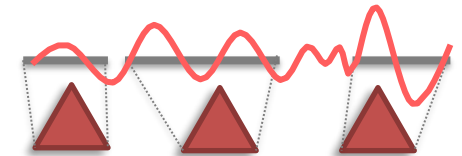


Definition: Model alignment between modalities with continuous signals and no explicit elements

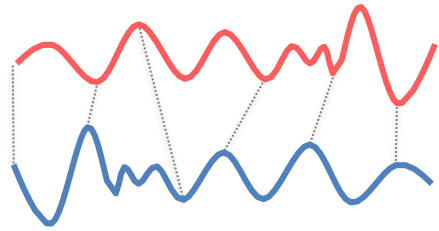
Continuous
warping



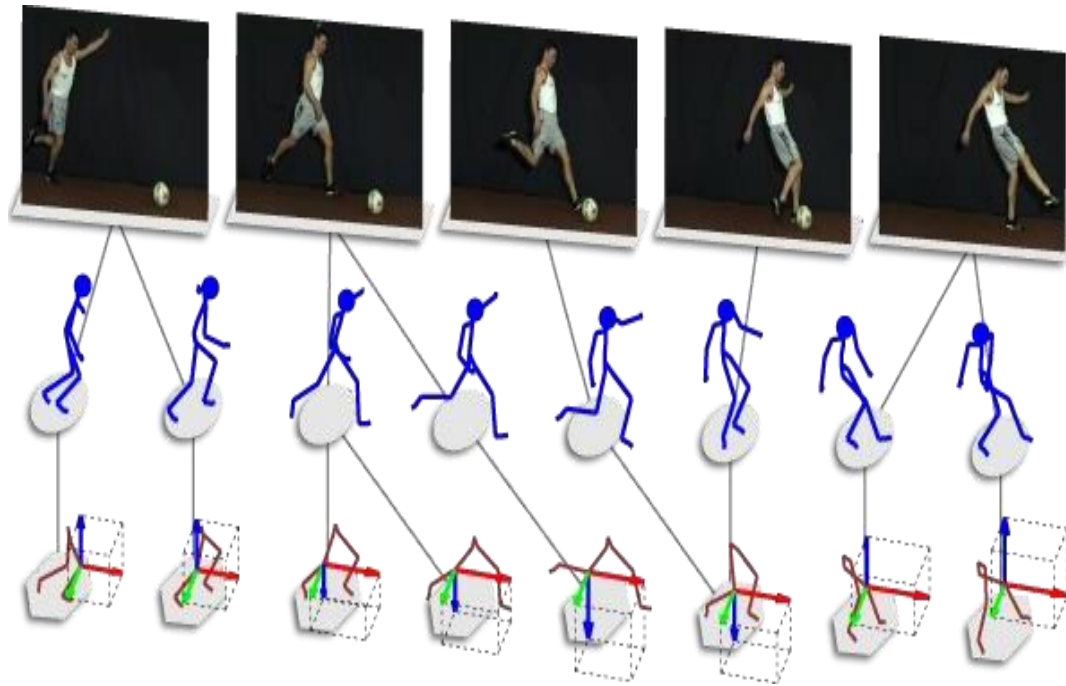
Discretization
(segmentation)



Continuous Warping – Example



➡ Aligning video sequences



Dynamic Time Warping (DTW)

We have two unaligned temporal unimodal signals

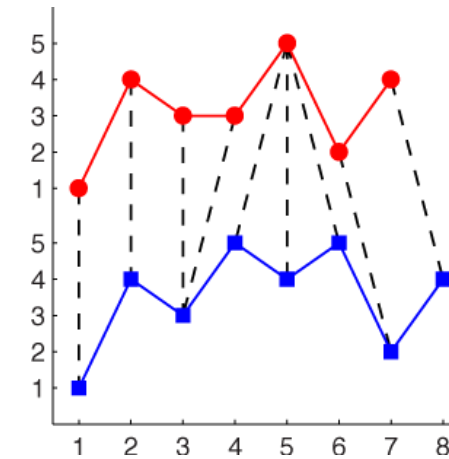
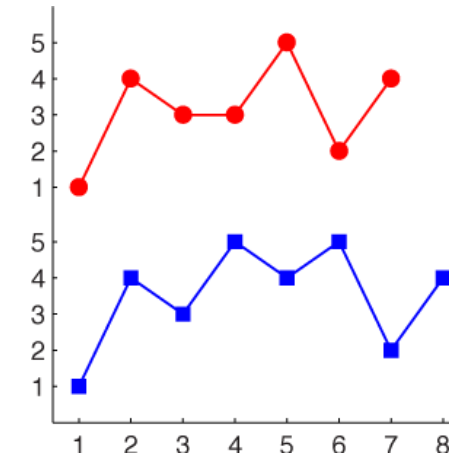
- $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_x}] \in \mathbb{R}^{d \times n_x}$
- $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n_y}] \in \mathbb{R}^{d \times n_y}$

Find set of indices to minimize the alignment difference:

$$L(\mathbf{p}^x, \mathbf{p}^y) = \sum_{t=1}^l \|\mathbf{x}_{p_t^x} - \mathbf{y}_{p_t^y}\|_2^2$$

where $\mathbf{p}^x$ and $\mathbf{p}^y$ are index vectors of same length

Dynamic Time Warping is designed to find these index vectors!



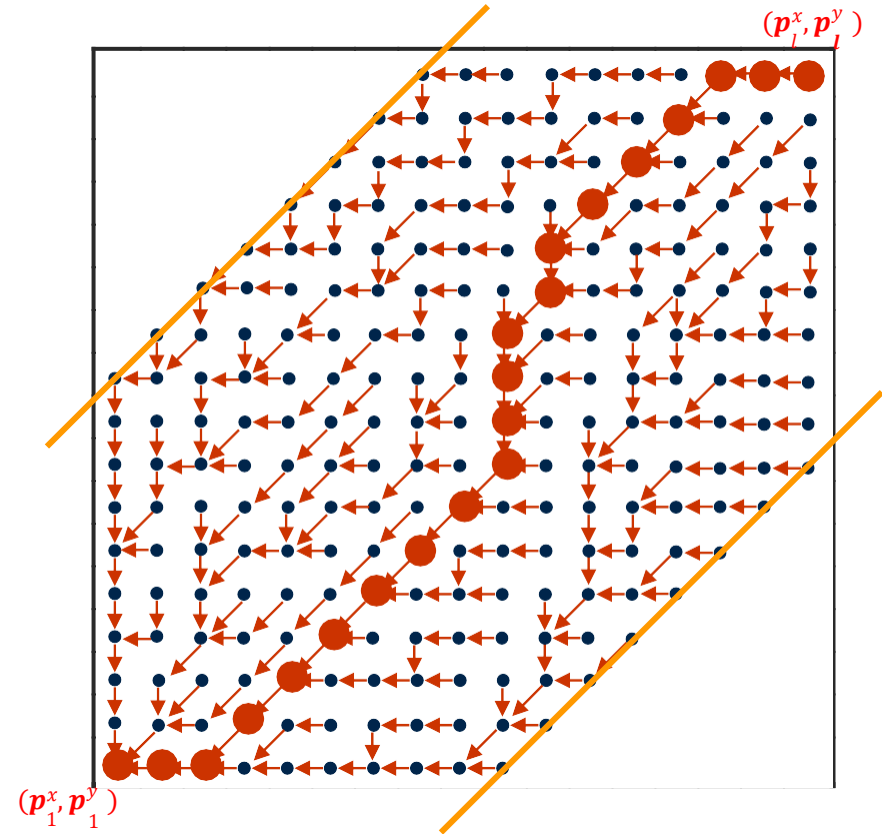
Dynamic Time Warping (DTW)

Lowest cost path in a cost matrix

- Restrictions?

- Monotonicity – no going back in time
- Continuity - no gaps
- Boundary conditions - start and end at the same points
- Warping window - don't get too far from diagonal
- Slope constraint – do not insert or skip too much

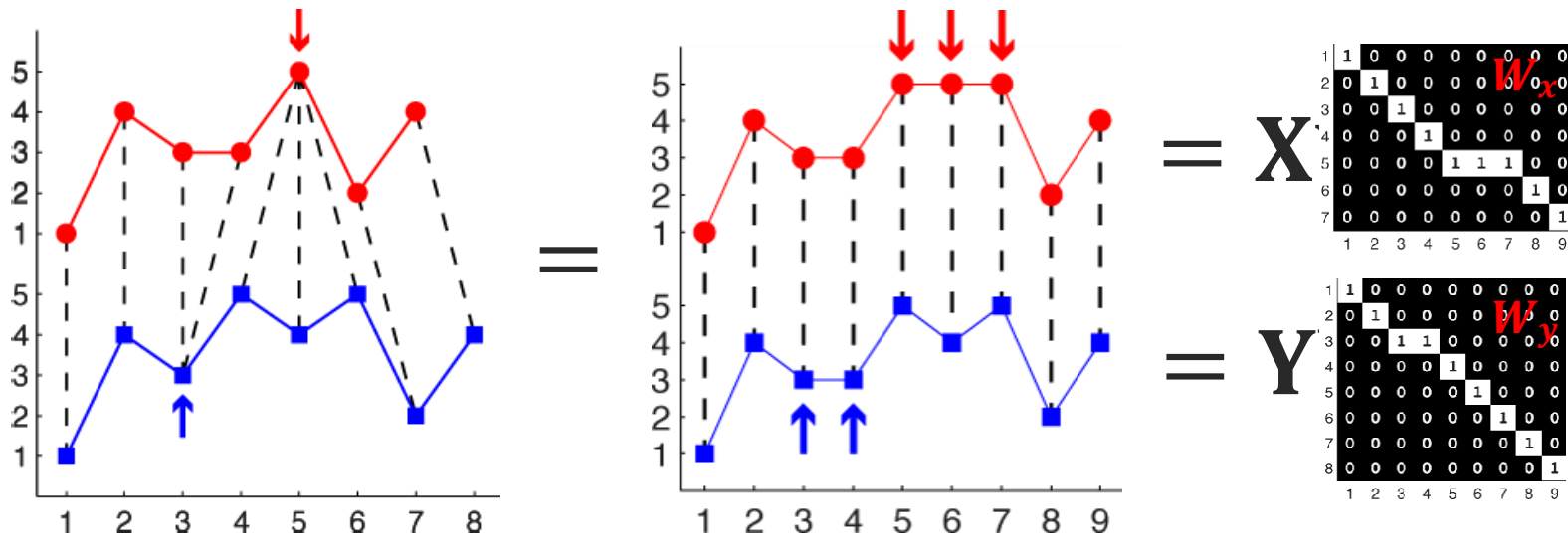
Solved using dynamic programming while respecting the restrictions



DTW alternative formulation

$$L(\mathbf{p}^x, \mathbf{p}^y) = \sum_{t=1}^l \left\| \mathbf{x}_{p_t^x} - \mathbf{y}_{p_t^y} \right\|_2^2$$

Replication doesn't change the objective!



Alternative objective:

$$L(\mathbf{W}_x, \mathbf{W}_y) = \left\| \mathbf{X}\mathbf{W}_x - \mathbf{Y}\mathbf{W}_y \right\|_F^2$$

Frobenius norm $\|\mathbf{A}\|_F^2 = \sum_i \sum_j |a_{i,j}|^2$

$\mathbf{X}, \mathbf{Y}$ – original signals (same #rows, possibly different #columns)

$\mathbf{W}_x, \mathbf{W}_y$ – alignment matrices

A differentiable version of DTW also exists...

<https://arxiv.org/pdf/1703.01541.pdf>

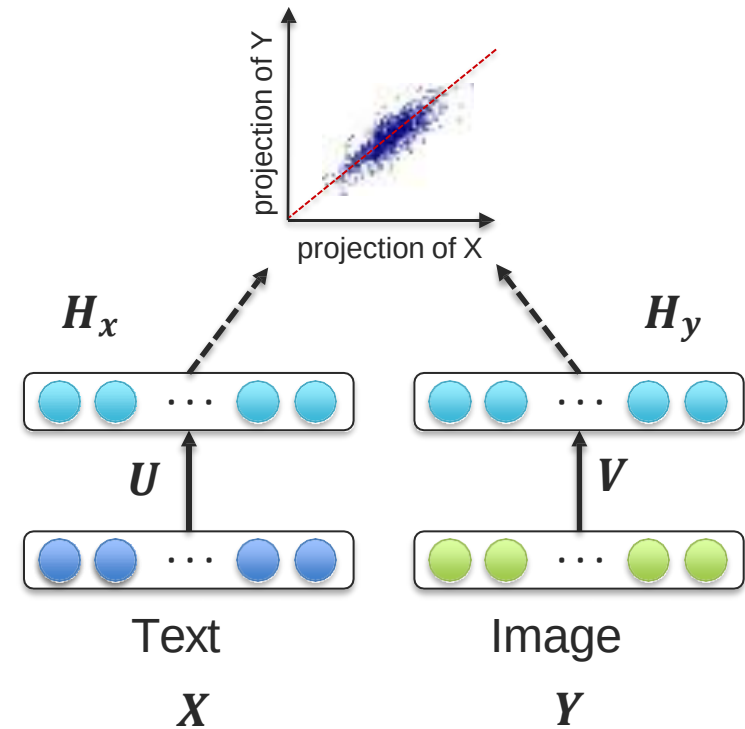
Canonical Correlation Analysis – Reminder

CCA loss can also be re-written as:

$$L(U, V) = \|\mathbf{U}^T \mathbf{X} - \mathbf{V}^T \mathbf{Y}\|_F^2$$

subject to:

$$\mathbf{U}^T \Sigma_{YY} \mathbf{U} = \mathbf{V}^T \Sigma_{YY} \mathbf{V} = \mathbf{I}, \mathbf{u}_{(j)}^T \Sigma_{XY} \mathbf{v}_{(i)} = 0$$



Canonical Time Warping

Dynamic Time Warping + Canonical Correlation Analysis = Canonical Time Warping

$$L(\mathbf{U}, \mathbf{V}, \mathbf{W}_x, \mathbf{W}_y) = \|\mathbf{U}^T \mathbf{X} \mathbf{W}_x - \mathbf{V}^T \mathbf{Y} \mathbf{W}_y\|_F^2$$

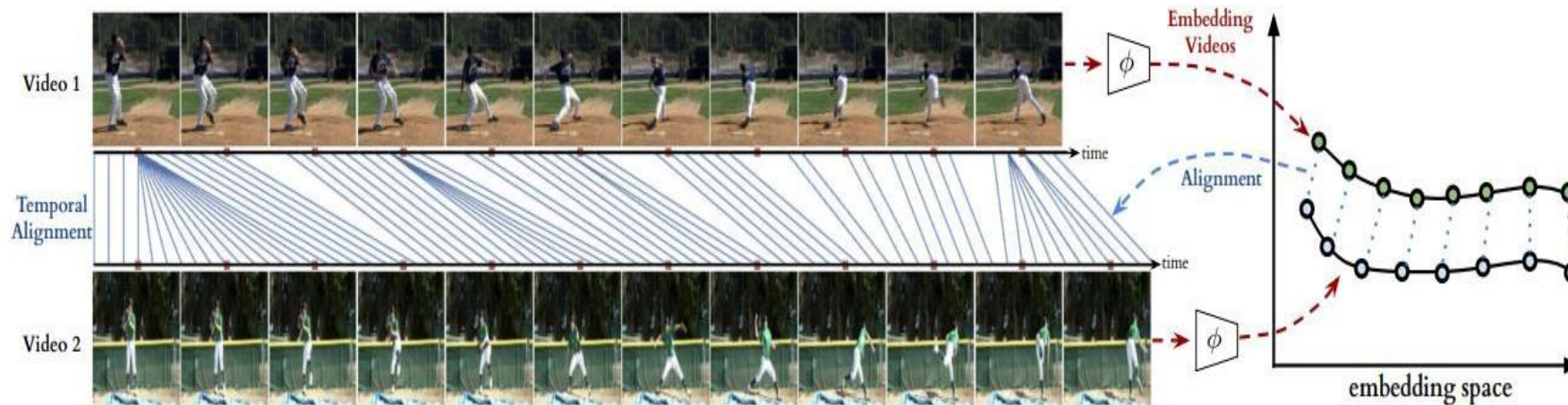
Allows to align multi-modal or multi-view (same modality but from a different point of view)

- $\mathbf{W}_x, \mathbf{W}_y$ – temporal alignment
- $\mathbf{U}, \mathbf{V}$ – cross-modal (spatial) alignment

[Canonical Time Warping for Alignment of Human Behavior, Zhou and De la Tore, 2009]

Temporal Alignment and Neural Representation Learning

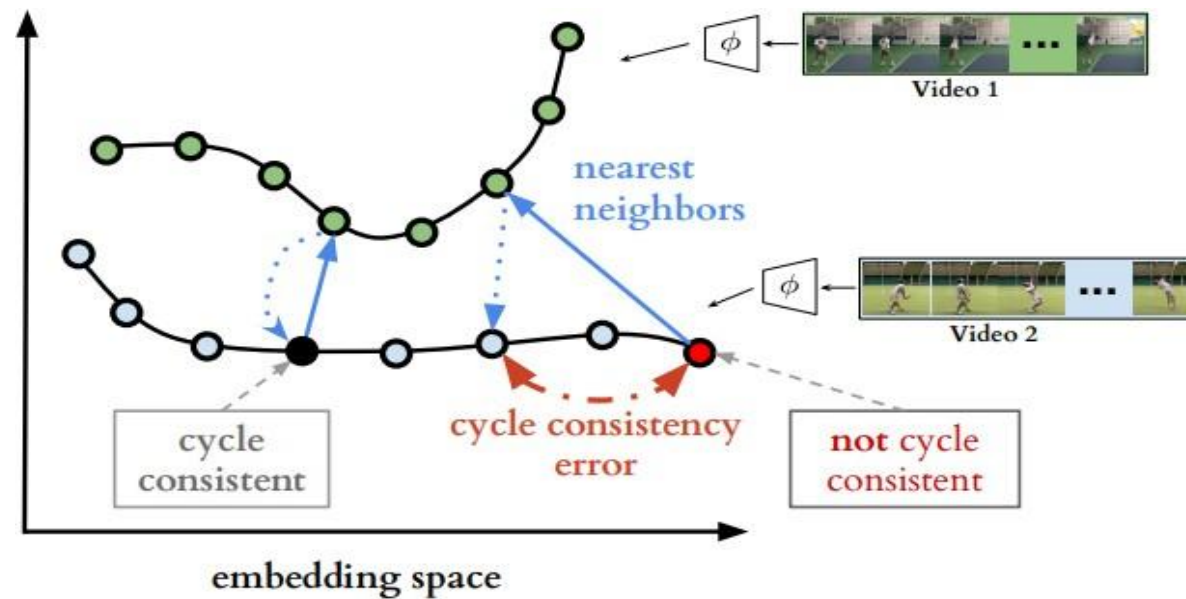
Premise: we have paired video sequences that can be temporally aligned



How can we define a loss function to enforce the alignment between sequences while at the same time learning good representations?

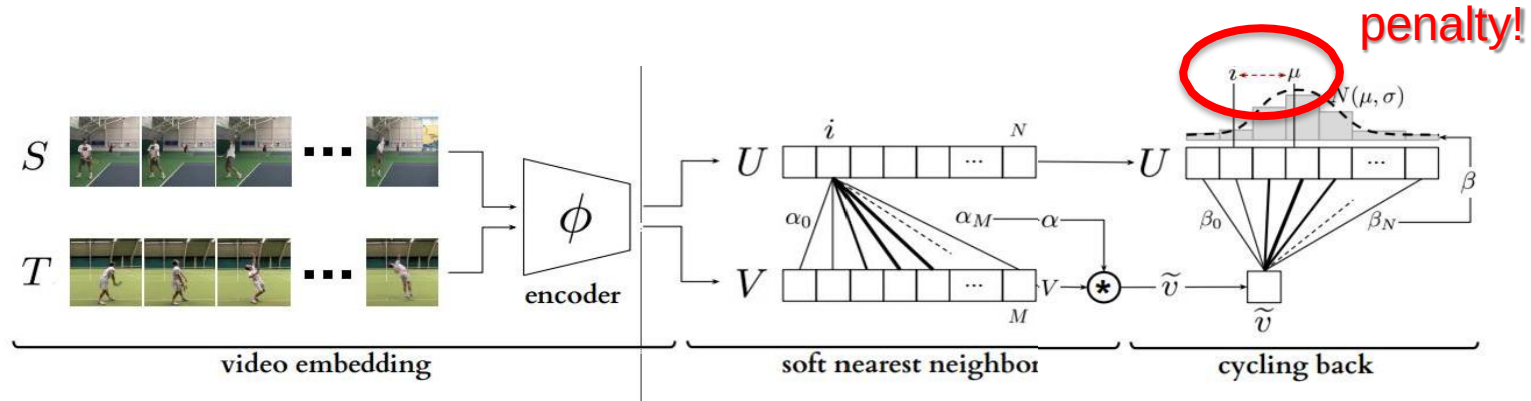
Temporal Cycle-Consistency Learning

Solution: Representation learning by enforcing **Cycle consistency**



Main idea: My closest neighbor also views me as their closest neighbor

Temporal Cycle-Consistency Learning



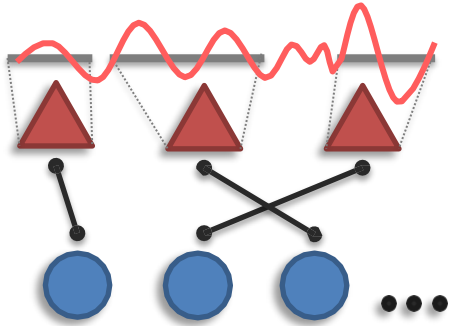
Compute “soft” / “weighted” nearest neighbour:

distances: $\alpha_j = \frac{e^{-||u_i - v_j||^2}}{\sum_k^M e^{-||u_i - v_k||^2}}$ Soft nearest neighbor: $\tilde{v} = \sum_j^M \alpha_j v_j,$

Find the nearest neighbor the other way and then penalize the distance:

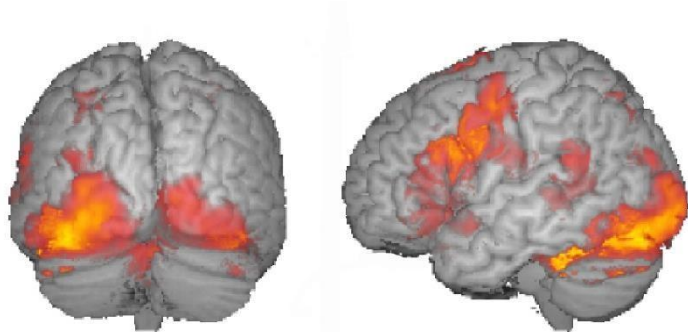
$$\beta_k = \frac{e^{-||\tilde{v} - u_k||^2}}{\sum_j^N e^{-||\tilde{v} - u_j||^2}} \quad L_{cbr} = \frac{|i - \mu|^2}{\sigma^2} + \lambda \log(\sigma)$$

Discretization (aka Segmentation)

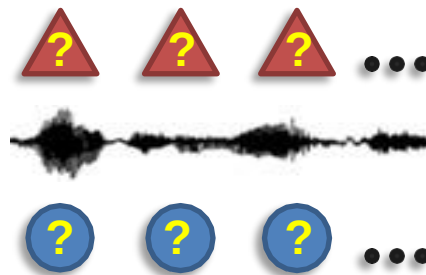


Common assumptions: ① Segmented elements

Examples:



Medical imaging



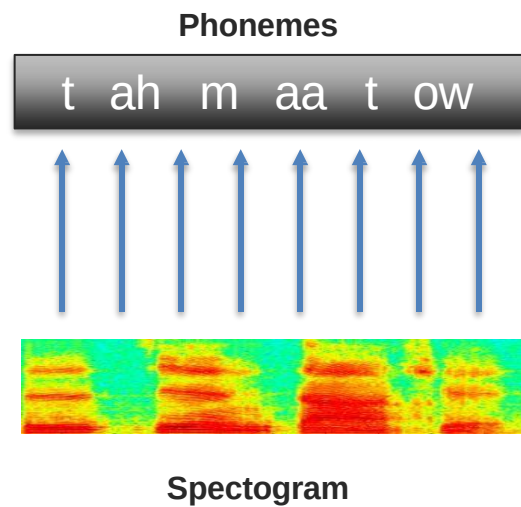
Signals



Images

Discretization – Example

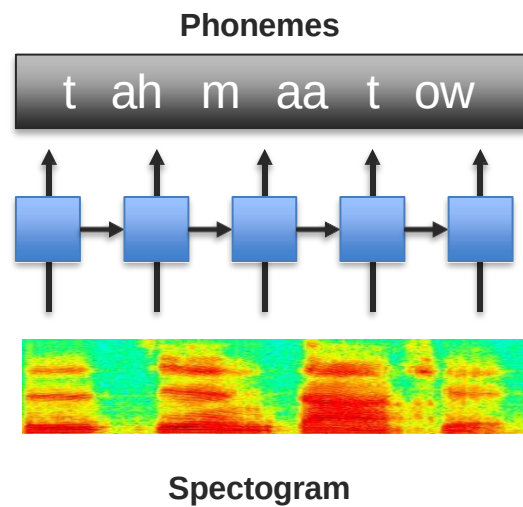
Sequence Labeling and Alignment



How can we predict the sequence
of phoneme labels?

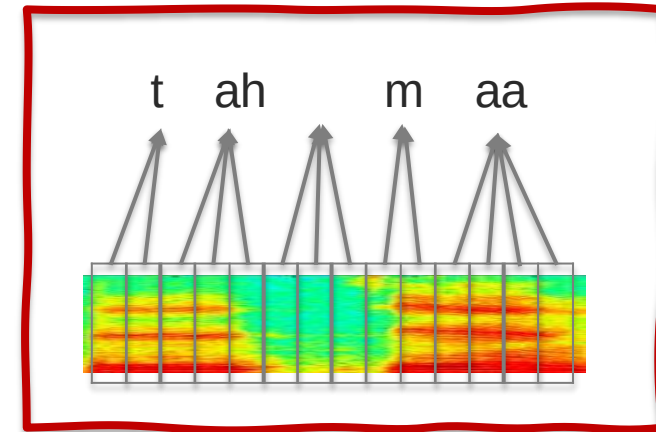
Discretization – Example

Sequence Labeling and Alignment



How can we predict the sequence
of phoneme labels?

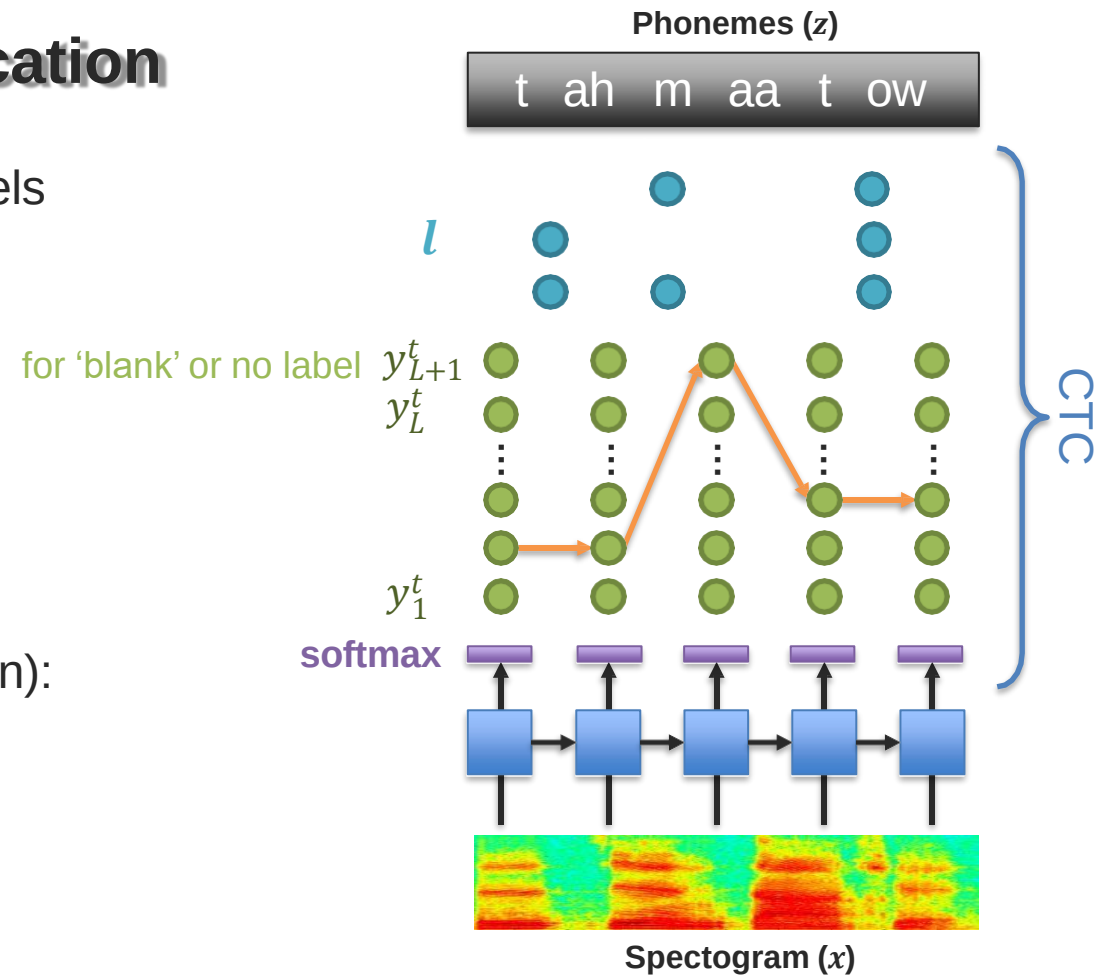
Challenge: many-to-1 alignment



Discretization – A Classification Approach

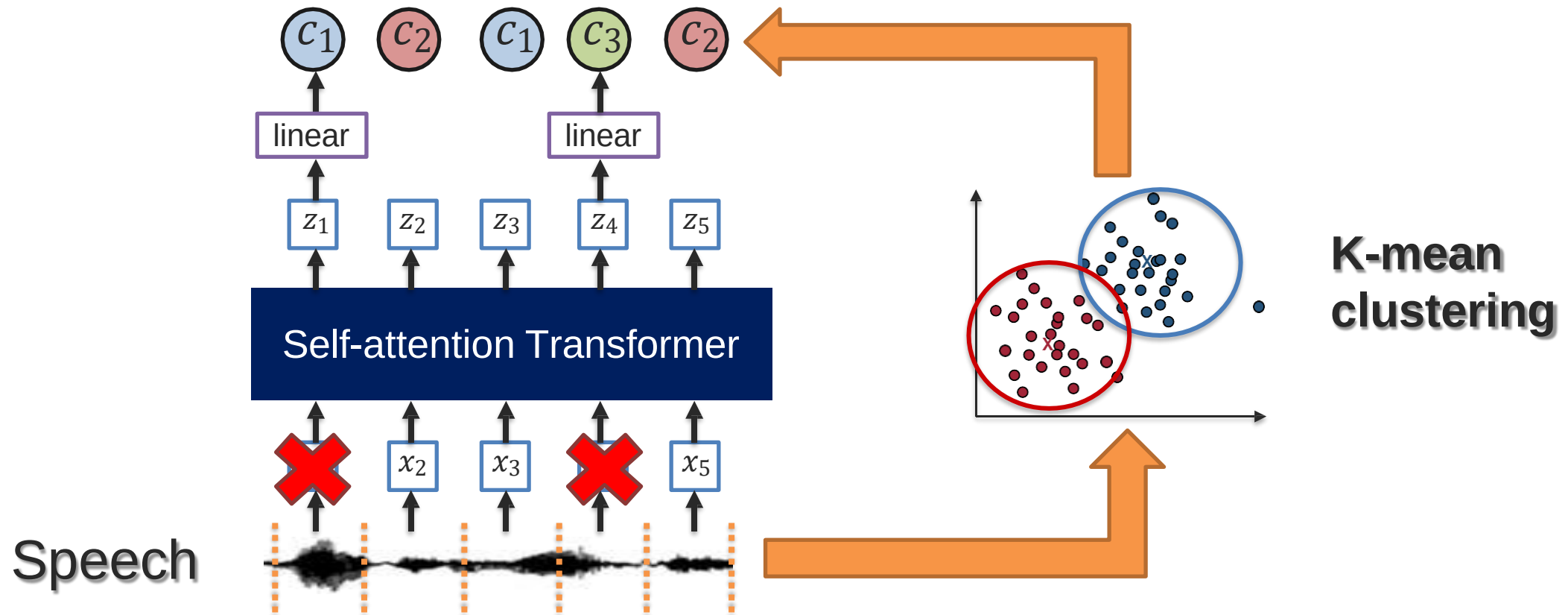
Connectionist Temporal Classification

- ④ Most probable sequence labels
- ③ Predicted labels l
- ② Path π over the activations:
- ① Output activations (distribution):



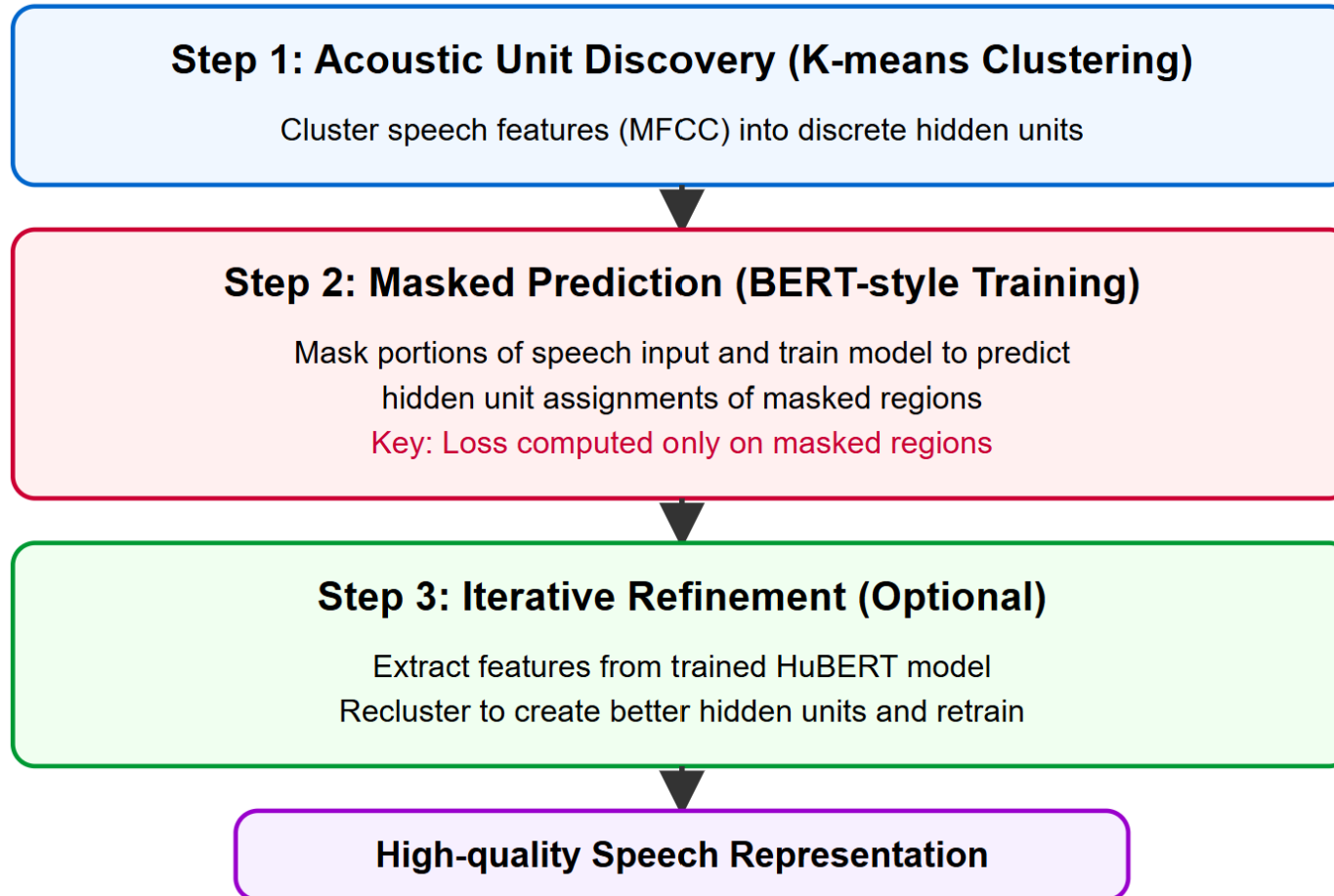
Discretization and Representation – Cluster-based Approaches

HUBERT: Hidden-Unit BERT



Discretization and Representation – Cluster-based Approaches

HuBERT: Hidden-Unit BERT Approach

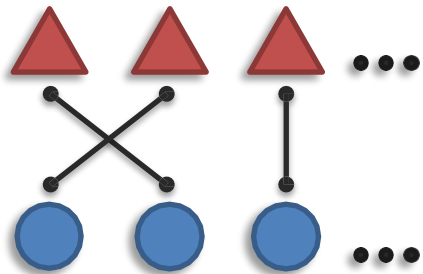


Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

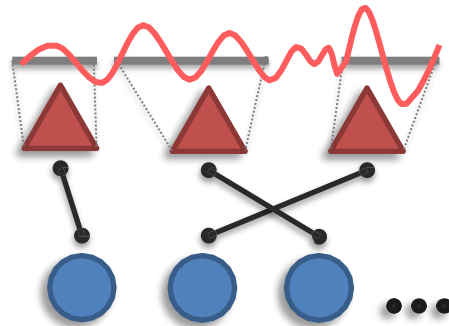
Sub-challenges:

Discrete Alignment



Discrete elements
and connections

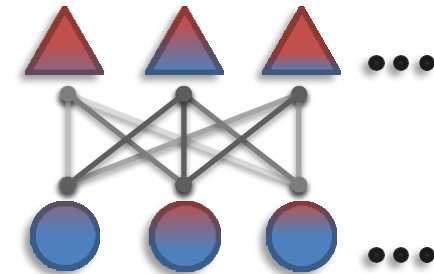
Continuous Alignment



Segmentation and
continuous warping

Next!

Contextualized Representation



Alignment
+ representation